Lab 1 – AI^2 Product Description

Craig Grubb

Old Dominion University

CS411W

Dr. Sarah Hosni

11 March 2026

Table of Contents

1 Introduction.....3

2 Product Description..... 5

2.1 Key Product Features and Capabilities.....5

2.2 Major Components – Hardware/Software..... 8

3 Identification of Case Study.....9

4 Glossary..... 11

5 References.....12

List of Figures

Figure 1: Current ADS Invoice Matching Workflow..... 4

Figure 2: Major Functional Components Diagram.....8

List of Tables

Table 1: Features and Capabilities..... 7

1 Introduction

Many organizations handle tens of thousands of invoices annually, predominantly through manual processes that create inefficiencies and risk. Manually handling those invoices can cost between twelve and thirty-five dollars per invoice. This is two to three times the cost of automating that process (Tamaro, 2025).

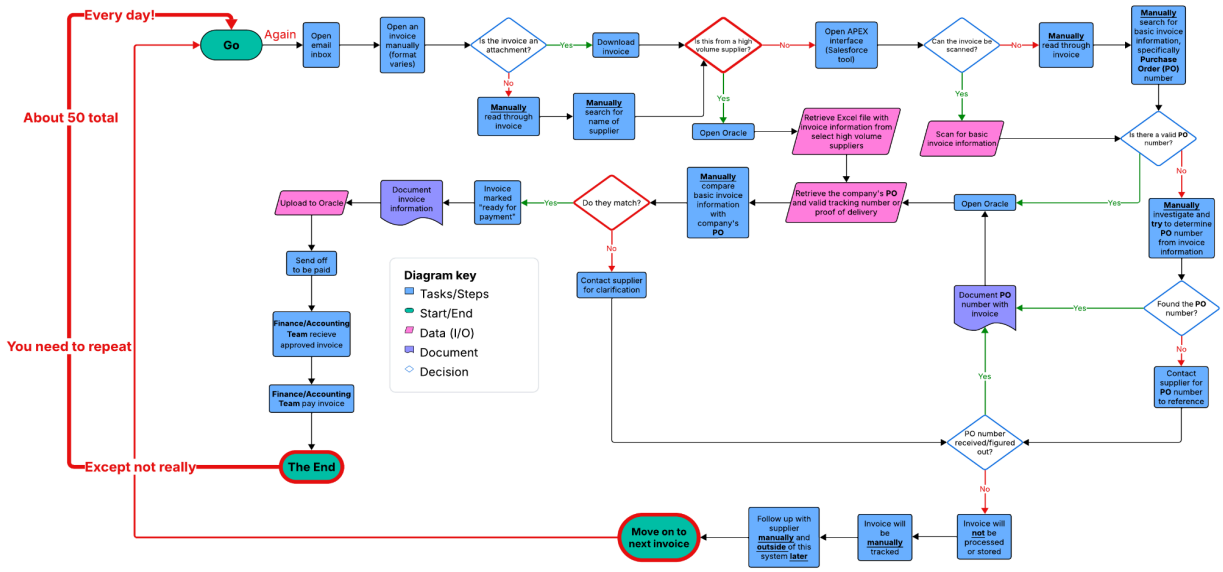
Manually matching incoming invoices to existing purchase orders leads to delays, data entry errors, and backlogs across accounting departments. A computer-automated system is better suited to this task. According to Megan O'Brien, content strategist for Oracle, automating financial processes can reduce errors, fraud, time and costs while increasing visibility, efficiency, and collaboration (2024).

To illustrate the scope of this problem, Atlantic Diving Supply serves as a representative case study. ADS receives around 150,000 invoices annually from roughly three thousand suppliers (Hosni, 2025).

To handle this volume, ADS employs a six-person accounts payable team. Each member of that team processes roughly fifty invoices per member each day. This manual process creates volume pressure with roughly six hundred invoices processed daily, leads to errors such as duplicate payments and PO mismatches, causes delays in supplier payments, and places excessive stress on AP staff. An outline of ADS' current workflow is illustrated on the next page in figure 1.

Figure 1

Current ADS Invoice Matching Workflow



A solution would not only help the ADS AP team, it would also help the suppliers receive payment more quickly, and it would help ADS management track financial performance.

The solution for these problems is AI² (Artificially Intelligent Invoices), a web-based intelligent automation system for invoice management.

AI² reduces manual workload, accelerates invoice turnaround, bolsters supplier satisfaction, and improves data accuracy with AI-driven recommendations.

AI^2 leverages artificial intelligence to match invoices to POs with confidence scoring, giving companies the insight on what invoice-PO matches need reviewing, which in turn improves model performance.

AI^2 improves efficiency, accuracy, adaptability, transparency, and scalability over manual processing. An API that pulls invoices from Emails, another that pulls PO information from a company's ERP, and a third that places the matched invoice into the ERP improves efficiency from end to end. The automation also removes human error to improve accuracy. The accuracy also improves over time with the model training inherent in the process, adapting the model to the company's unique challenges. Confidence scores and logs provide end users the transparency needed to trust the software is making appropriate decisions. Finally, automation makes the process scalable, eschewing the need for additional staffing as volume increases.

2 Product Description

AI^2 is a web application that leverages artificial intelligence models when a simple if…then statement falls short in matching an incoming invoice with a purchase order.

2.1 Key Product Features and Capabilities

AI^2 is an end-to-end solution, from invoice intake to sending the match to an ERP solution.

Any secured application begins with authentication. AI^2 works with a company's existing SSO or LDAP solution to make the process as seamless as possible.

Once logged in, a member of the AP team can access their assignment queue where they can see what matches need their review. Managers can log in to manually assign invoices or allow the system to automatically assign work. AP team members can place themselves in or remove themselves from the queue – or a manager can manage team members in the queue.

Managers and administrators can access a summary dashboard to view performance and audit logs. There, they can also access the confidence scores of each match.

If a match does have a low confidence score, the results of the human review is fed back into the AI model so that the application can learn the edge cases the company faces to improve matching reliability.

To further enhance efficiency, AI^2 has an API to pull invoices directly from an Email inbox, beginning the process. This pushes the invoice to a programmatic algorithm to see if the invoice can be immediately matched to an invoice.

[This space intentionally left blank.]

Table 1

Features and Capabilities

Feature	**Description**
Authentication	Integrates with a company's SSO or LDAP solution.
Assignment Queue	Match reviews can be assigned automatically or manually.
Summary Dashboard	Interface for performance and audit logs.
Confidence Scoring	Provides transparency on the model's confidence of a match.
Continuous Learning	Feeding manual matches into the AI model allows companies to cover edge cases.
Email Intake	Pulls invoice directly from the inbox.
PO Matching API	Matches invoices to existing POs with AI when they cannot be programmatically matched and enters those matches into the company's ERP.
Audit Logging	Provides administrators insights into how the software is matching invoices to POs.
Performance Logs	Gives management data to evaluate system performance and allows administrators to judge the efficacy of different models.

If not, the invoice is then fed to the PO matching API, where artificial intelligence is used to evaluate the invoice against existing POs to find a match. The AI assigns a confidence score.

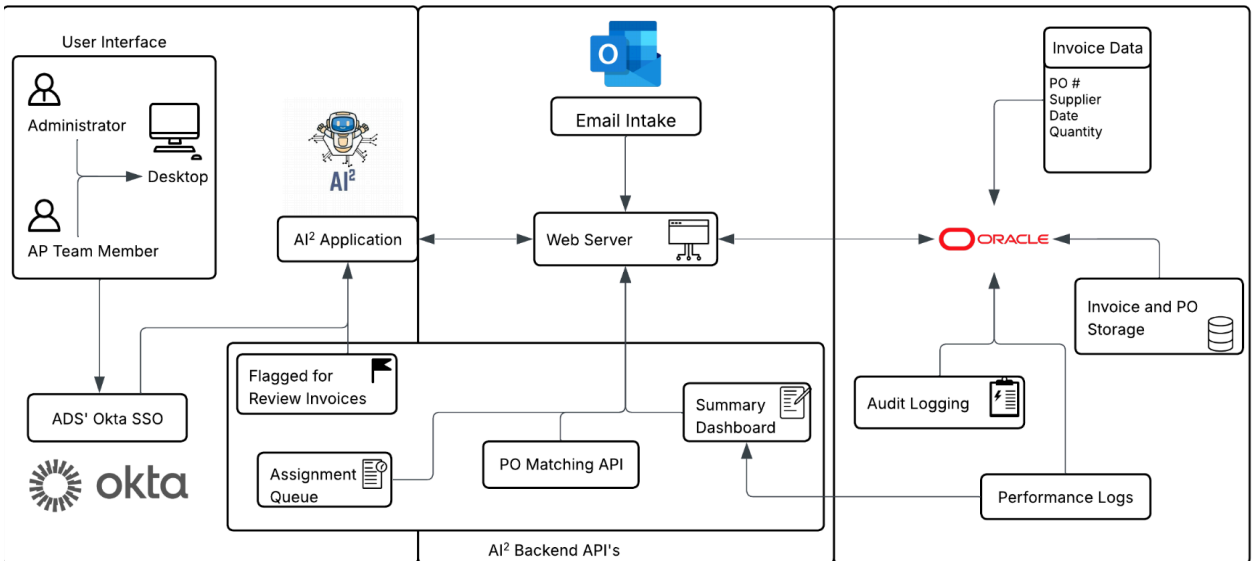
If the confidence score is high enough, the match is automatically fed to the company's ERP. Otherwise, it is added to the assignment queue. Table 1 provides an outline of AI²'s features and capabilities.

2.2 Major Components – Hardware/Software

AI² is a web application. Therefore, it can be accessed from any computer running a modern web browser. The user interface will vary depending on the role a user may have, removing unnecessary clutter while retaining the features needed to accomplish their job. Figure 2 shows a diagram of the major functional components of AI².

Figure 2

Major Functional Components Diagram



Once a member has logged in through Okta SSO, they will see dashboards that show their assignment queue, invoices flagged for review, and purchase orders that the PO matching API will suggest to match invoices to.

Administrators will login to access performance logs, audit logs, and a summary dashboard.

Bookending this entire process is the Email intake API that will automatically ingest incoming invoices and an ERP API that gathers purchase orders and saves its corresponding matched invoice. These APIs further time savings by automating tasks that would take much longer if done manually.

AI^2 stores its own information on matches, performance, and AI models in a MariaDB database. The web application itself leverages a monolithic architecture built using Flask.

3 Identification of Case Study

For the sake of an example, there is a logistics firm called Monarch Shipping in Norfolk. In that company, there are four AP clerks, an AP manager, and a systems administrator.

This AP team handles 225 invoices each day. The team is stretched thin, and the AP manager is working on a business case to invest in an automated system rather than encumber the company with another salary.

With AI^2, the AP manager could cover his labor shortage, making his team much more efficient. It would also give them greater insight into what their employees are accomplishing on a daily basis and what kinds of challenges they are facing.

The AP team would benefit from more interesting work. Rather than engaging in simple, repetitive tasks, the team members could simply review AI-recommended matches and concentrate on the more challenging matches. Even the more challenging matches would be simpler, though, as AI^2 would have cleared the backlog of purchase orders, giving the team fewer to search when looking for a match.

Finally, adding software is going to add work to IT staff. AI^2 needs to provide simple and intuitive dashboards and configurations to ease that burden. Automated document processing also helps with the sysadmin's digital transformation initiative, which many companies are struggling with (Mazza, n.d.)

4 Glossary

Application Programming Interface (API): an interface a program provides to another to provide a specific service.

Enterprise Resource Planning (ERP): software, or different software integrated together, that consolidates and interprets data to empower business to make better decisions.

Invoice: a bill for goods or services issued to a customer after the customer has issued a purchase order.

Purchase Order (PO): a written request for goods and services with an agreement to pay a price previously provided by the business in the form of a written quote.

5 References

Mazza, J. (n.d.). *Four Benefits of Applying AI-Led Automation to Document Processing*. IBM. Retrieved March 11, 2026, from, https://www.ibm.com/think/insights/four-benefits-of-applying-ai-led-automation-to-document-processing

O'Brien, M. (2024, July 30). *Finance Automation in Detail*. Oracle. https://www.oracle.com/erp/finance-automation/#:~:text=It%20allows%20for%20areas%20with,capabilities%20of%20finance%20automation%20forward.

Tamaro, R. (2025, March 4). *The Hidden Costs of Manual Accounts Payable and How Automation Solves Them*. CASO Document Management. https://caso.com/2025/03/the-hidden-costs-of-manual-accounts-payable-and-how-automation-solves-them/